Learning absolute protein stability

Accurate protein stability prediction for small domains using mega-scale experiments

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Journal Club

Can we predict whether a protein will stay folded?

A plausible structure does not establish its thermodynamic stability.

Existing strengths

Structure predictors estimate a folded conformation.

Mutation predictors often estimate the effect of a single substitution.

The missing quantity

Absolute folding stability of an arbitrary small domain, on an experimental energy scale.

Useful for ranking unrelated natural proteins and new designs.

Question: can diverse, uniform experimental labels turn a structure-aware language model into an absolute stability predictor?

Cho et al., Introduction; Figs. 1–2.

Define the target before comparing predictions

Here, stability means the free energy of *unfolding*:

$$\Delta G_{\mathrm{unfold}} = G_U - G_F = RT \ln \frac{p_F}{p_U}.$$

Higher positive values mean the folded state is more favored.

Absolute stability
ΔG: one sequence at specified conditions.

Training labels: approximately -1 to $5\,\mathrm{kcal\,mol^{-1}}$, pH 7.4, 298 K.

Mutation effect
$\Delta\Delta G = \Delta G_{\mathrm{mutant}} - \Delta G_{\mathrm{WT}}$.

With this sign convention, positive $\Delta\Delta G$ means *stabilizing*.

Paper, Introduction and Methods. Equation is the two-state thermodynamic definition.

The contribution is diversity, not just more mutants

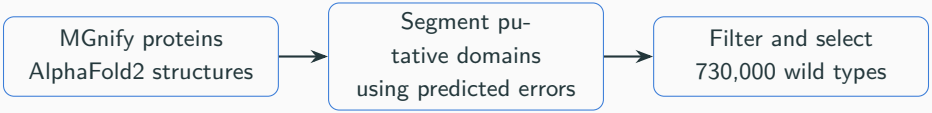
	Earlier Megascale	MGnify Stability
Experimental sequences	776,000	$>$ 1.8 million
Sequence clusters (30% identity)	199	211,020
Structural clusters (TM-score 0.5)	—	129,303
Main sampling strategy	Mutants of few domains	Diverse domains + variants

The earlier dataset sampled local sequence neighborhoods deeply. The new dataset broadens the number of domain families with measured absolute stability.

Why this matters
Learning how mutations change one fold need not teach the stability baseline of a different fold.

Introduction; Fig. 1a–c. Earlier Megascale contains 479 WT domains before 30%-identity clustering.

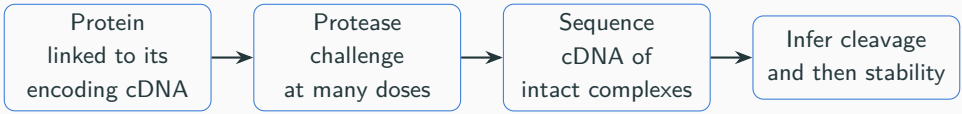
What was sampled, and what was excluded?



- Domains are 60–80 amino acids, with mean pLDDT ≥ 50 and secondary structure; cysteine-containing domains are removed in library design.
- Add 710,605 single substitutions, 150,095 double substitutions, 1,889 higher-order variants, and 292,486 indels.
- Highly hydrophobic sequences are later excluded from machine learning because they can appear spuriously stable.

Fig. 1c; Methods: library design and clustering. Counts describe library composition, not final training size.

How a pooled assay becomes an energy measurement



Key inference: compare measured protease susceptibility with a calibrated prediction for the same sequence in its unfolded state.

Assumption to watch
Resistance reflects equilibrium folding protection. Cleavage from a folded loop, nucleic-acid binding, or unusual unfolded-state cleavage can bias the inferred ΔG.

Results: measuring stability; Methods: cDNA display proteolysis; underlying protocol Tsuboyama et al., Nature 2023.

Independent measurements support the assay scale

Validation	Reported agreement
Trypsin versus chymotrypsin	RMSE 0.88; 0.70 after global-offset removal; Pearson $r = 0.89$
1,997 shared sequences across libraries	RMSE 0.36 after calibration
20 individually purified domains: chemical denaturation monitored by CD	RMSE 1.00; Pearson $r = 0.86$
Same purified set, omitting one outlier	RMSE 0.79; Pearson $r = 0.91$

All RMSE values are in $\mathrm{kcal\,mol^{-1}}$. The purified-protein check validates the physical scale but samples only 20 domains.

Fig. 1e; main-text descriptions of Supplementary Figs. 1–2. Offset correction and outlier removal are explicit.

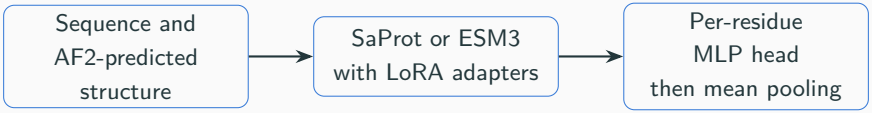
Training and testing are separated by sequence families

Final machine-learning split	Sequences
Training: 525,127 WT + 435,089 substitution variants	960,216
Validation	2,357
Test: 590 sequence clusters	3,283

- Randomly assign 30%-identity clusters; remove detectable training–evaluation similarity with JackHMMER.
- No indels in the primary training set: indels test a new variant class.
- Structural separation is incomplete: 1,590 test structures cluster with training at TM-score 0.5. Reported errors remain similar across similarity strata.

Methods: library design and clustering; Supplementary Fig. 6 discussed in main text. Pretraining exposure is a separate issue.

Sequence + structure ⟶ one stability number



$$\widehat{\Delta G}_{\mathrm{raw}} = \frac{1}{L}\sum_{i=1}^{L} h_\theta(z_i), \qquad W = W_0 + BA.$$

- SaProt: 650M model, 20 structural tokens; ESM3: 1.4B model, 4,096 structural tokens.
- LoRA rank 4; about 0.72% and 0.295% of backbone parameters updated, respectively. Final predictors ensemble three runs.

Per-residue head outputs are pooled latent predictions, not experimentally resolved energetic contributions.

Fig. 2c; Methods: fine-tuning absolute stability models.

A concrete example explains the joint objective

Illustrative values, not an experimental protein:

	WT ΔG	Mutant ΔG	$\Delta\Delta G$
Experimental target	2.0	3.0	+1.0
Model prediction	2.5	3.1	+0.6

$$\mathcal{L} = 0.3(\widehat{\Delta G}_{\mathrm{WT}} - \Delta G_{\mathrm{WT}})^2 + 0.3(\widehat{\Delta G}_{\mathrm{MT}} - \Delta G_{\mathrm{MT}})^2 + (\widehat{\Delta\Delta G} - \Delta\Delta G)^2.$$

For these values, loss $= 0.3(0.5)^2 + 0.3(0.1)^2 + (-0.4)^2 = 0.238$.

Why both terms?
Absolute labels anchor the energy baseline; the difference term explicitly teaches mutation effects. Good ΔG predictions alone need not yield good $\Delta\Delta G$.

Methods: loss function with $\Delta\Delta G$ regularization. Toy example constructed for teaching.

Assay censoring and terminal cleavage change the model

Range-corrected head
Linear for raw scores 0–4; sigmoid tails approach the assay bounds -1 and 5.

Use this correction for cDNA labels; bypass it for conventional stability measurements.

Bypassing a cap enables extrapolation; it does not validate extrapolation.

Augmented variants
Unstructured termini may be cleaved even in folded proteins, reducing apparent stability.

Filter to a 456,000-sequence subset with structured termini; randomly perturb terminal segments during training.

Augmented ESM3 improves transfer to conventional assays, while worsening assay-matched test accuracy.

Methods: sigmoid-corrected head and terminus augmentation; main-text descriptions of Supplementary Figs. 3–4.

Main result: experimental fine-tuning adds a large signal

Predictor on the 3,283-sequence MGnify test	Spearman ρ	RMSE
Buried nonpolar area from hydrophobic residues	0.60	—
ESM-IF sequence-likelihood score	0.43	—
AlphaFold2 pLDDT	−0.17	—
SaProtΔG	**0.88**	**0.80**
ESM3ΔG	**0.87**	**0.80**

RMSE in $\mathrm{kcal\,mol^{-1}}$. These are sequence-family-held-out predictions within the assay's energy and domain-size range.

Interpretation
Structure confidence and evolutionary likelihood are useful inputs, but do not directly measure thermodynamic stability. The negative pLDDT result is conditional on the library's preselection.

Fig. 2a–b; Results: existing models and fine-tuning. Use exact main-text values rather than abstract rounding.

The model learns more than hydrophobic burial alone

Restrict evaluation to 2,006 domains with buried nonpolar area $> 3{,}000\,\text{Å}^2$.

Spearman correlation with stability	Full test	High-burial subset
Buried hydrophobic nonpolar surface area	0.60	0.36
SaProtΔG	0.88	0.80
ESM3ΔG	0.87*	0.77

Sequence-likelihood scores lose almost all correlation in this subset, according to the main text. **What this establishes:** predictions contain useful variation beyond one dominant structural feature. It does not identify a mechanistic decomposition of folding forces.

Main-text description of Supplementary Fig. 5. *The local paragraph rounds ESM3's full-test value to 0.88.

Absolute predictions recover point-mutation effects

28,172 substitutions from 28 WT proteins	SaProtΔG	ESM3ΔG
$\Delta\Delta G$ RMSE ($\mathrm{kcal\,mol^{-1}}$)	0.76	0.71
Stabilizing-versus-destabilizing ROC AUC	0.94	0.94
Matthews correlation coefficient	0.58	0.60

ThermoMPNN: AUC 0.93, MCC 0.59; overall regression performance is comparable.

- Classification uses 1,178 stabilizing and 12,309 destabilizing variants; stabilizing threshold $\Delta\Delta G > 0.5\,\mathrm{kcal\,mol^{-1}}$.
- Nine WT proteins overlap training above 30% identity. Results remain similar for the other 19 (21,118 variants).

Fig. 3b–c; main-text descriptions of Supplementary Figs. 7–8. All mutation signs follow unfolding convention.

Indels show why an absolute predictor is useful

Evaluate each sequence independently

WT structure + sequence $\rightarrow \widehat{\Delta G}_{\mathrm{WT}}$

Insertion/deletion structure + sequence $\rightarrow \widehat{\Delta G}_{\mathrm{variant}}$

Subtract predictions to obtain $\widehat{\Delta\Delta G}$.

No fixed substitution site or equal-length assumption is needed.

106,210 MGnify indels

Not used in primary training.

RMSE: 0.79 (SaProt), 0.81 (ESM3) $\mathrm{kcal\,mol^{-1}}$.

91 known stabilizing indels

ESM3 correctly classifies 83.7%, versus 70.4% reported for Rosetta.

This selected positive set measures recovery of stabilizing effects, not population-wide accuracy.

Fig. 3d–f; Methods: indel quality filters; Gutierrez and Rocklin stabilizing-indel benchmark.

Transfer to larger, highly stable proteins is uneven

S1724: 1,724 conventional measurements from 35 WT proteins; reversible two-state unfolding, pH 5–8, 15–30 °C.

Promising evidence

Augmented ESM3, with sigmoid disabled, improves over the base model.

Many domains longer than training show $\Delta\Delta G$ RMSE $< 2\,\mathrm{kcal\,mol^{-1}}$ and rank correlation > 0.7.

Clear failure boundary

Errors rise for large and highly stable domains, especially WT $\Delta G > 10\,\mathrm{kcal\,mol^{-1}}$.

Human growth hormone (3HHR): WT $\Delta G = 13.1\,\mathrm{kcal\,mol^{-1}}$; absolute predictions deviate substantially.

Benchmark statistics are evaluated within WT families; limited sample sizes produce wide uncertainty intervals.

Fig. 4; Methods: S1724 curation. Conditions differ from training; ThermoMutDB entries were manually checked/corrected.

Thermophile trends are a proxy, not direct calibration

External test: TED domains from 583 CATH families and 7,855 organisms; single-domain proteins up to 200 residues. Within each family, subtract the mean prediction for mesophilic organisms (20–30 °C), then compare predicted stability with optimal growth temperature.

Observed trend
Augmented ESM3 predicts higher stability for hotter-growing organisms in nearly all families. Pooled slope: $2.43\,\mathrm{kcal\,mol^{-1}}$ per 100 °C, steeper than the compared confidence/likelihood scores.

Interpretation: captures thermal-adaptation trends beyond training size. Organismal growth temperature is neither a protein's melting temperature nor its measured room-temperature ΔG.

Fig. 5 and Methods. ProteinMPNN/pLDDT were quantile-matched to the predicted-ΔG distribution before slope comparisons.

Protein-design tests support practical ranking

Evaluation	Result	What it supports
634 Rosetta designs, 12 studies; purified-protein folding	ESM3 AUC 0.76; pLDDT 0.65; MPNN 0.68	Transfer to a different assay and somewhat larger designs
20,572 small de novo designs; cDNA proteolysis	$\rho = 0.88$; RMSE 0.98 $\mathrm{kcal\,mol^{-1}}$	Strong generalization to designed sequences within assay regime
530 nanobodies; apparent melting temperature	Spearman $\rho = 0.65$	Ranking evidence across larger antibody scaffolds

Nanobody apparent melting can be irreversible; its association with predicted ΔG is not an absolute thermodynamic calibration.

Fig. 6a–f and main text. Rosetta AUC uncertainties reported as ± 0.02; dataset counts differ slightly within the paper.

Binder triage adds signal, with important qualifications

- Test designs from RFdiffusion's five targets and 302 de novo EGFR designs from the Adaptyv competition.
- Score the *binder–target complex*'s predicted folding stability. Successful binders tend to receive higher scores.
- ESM3 exceeds pLDDT and ProteinMPNN discrimination for most targets; insulin receptor is an exception. AF3 ipSAE generally performs better, except for TrkA.

What is actually being scored?
Complex folding stability is not binding free energy. Designs were already generated/filtered with ProteinMPNN and AF2, narrowing those baselines' score ranges.

Stability is a complementary ranking feature; prospective binder-design improvement remains to be established.

Fig. 6g–i; Methods: binder benchmark. Last sentence is our interpretation of retrospective evidence.

Where would we trust the prediction?

Strongest evidence
Compact, single small domains.

Assay-matched stability in the -1 to $5\,\mathrm{kcal\,mol^{-1}}$ range.

Ranking related variants, including indels.

Ranking small de novo designs.

Boundaries to test
Large, highly stable, disulfide-rich or unusual sequences; multiple linked domains.

Protease-sensitive folded termini; nucleic-acid binding can inflate stability.

Different environments; input-structure sensitivity and conformational ensembles.

Discussion: what independent, purified-protein test would separate a better folding predictor from a better predictor of this assay?

Discussion; Fig. 4; main-text discussion of Supplementary Fig. 10.

Take-home and reproducible starting points

Main advance: diverse experimental labels let structure-aware language models learn an absolute stability scale, with about 0.8 $\mathrm{kcal\,mol^{-1}}$ test RMSE for small domains. **What**

remains open: broad calibration beyond the training energy/size range and separation of folding physics from assay artifacts. Paper: doi.org/10.64898/2026.05.19.726285 (bioRxiv v1; May

20, 2026).

Full text: PMC13228446.

Code: yehlincho/absolute-stability-predictor.

Data: Zenodo 19411306; weights: Yehlin/absolute-stability.

Citation checked November 1, 2026; bioRxiv API lists no journal publication. Source/access and QA notes are in the adjacent README.